

Soham Nigam

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SUMMARY

Performance-driven AI Engineer and Software Developer specializing in Backend Development and scalable AI systems. Experienced in deploying low-latency Gemini-based agents, end-to-end MLOps pipelines, and translating Machine Learning research into production-ready AI infrastructure for advanced automation and predictive analytics.

SKILLS

Programming Languages: Python (Expert), Swift, JavaScript (ES6+), SQL, Solidity, Bash, C++.

AI & Machine Learning: Gemini 1.5 Pro, OpenAI GPT-4o, LangChain, RAG, ReAct Agents, NLP, Deep Learning, Computer Vision, Prompt Engineering, LLMs, Machine Learning.

Frameworks & Tools: FastAPI, React.js, Pydantic, Docker, MLflow, DVC, Git, REST APIs, LangGraph, Software Engineering, Full-Stack Development.

Backend & Cloud: Vector Databases (Pinecone), MongoDB, CI/CD, JWT Authentication, WebSockets, GCP, AWS (S3/EC2), Vertex AI, Model Deployment.

Specialized Domains: Google Maps Platform, MapKit, SwiftUI, Sentinel Hub, Polygon (Web3), IPFS, Data Processing.

EXPERIENCE

Stark Digital Media Services Pvt. Ltd.

Pune, India

Artificial Intelligence Intern

June 2025 – July 2025

- Engineered speech-to-speech NLU pipeline for real-time support, optimizing WebSocket data streams to achieve sub-500ms response latency.
- Developed scalable RAG system using Pinecone and Gemini, significantly improving document search efficiency across 500+ corporate assets.
- Orchestrated Computer Vision identity verification framework that automated validation workflows, reducing manual processing overhead.

PROJECTS

LabChain Sentinel – Gemini-Powered Autonomous Procurement Agent

- Architected closed-loop agentic system using ReAct framework to monitor laboratory inventory and autonomously trigger NABL-compliant purchase orders.
- Implemented strict Pydantic validation layers to enforce data contracts between LLM decision-making pipelines and FastAPI procurement backend.
- Designed automated logging system for audit trails, ensuring every autonomous procurement action meets rigorous regulatory compliance standards.
- Tech:** Python, FastAPI, Gemini API, Pydantic, React.

MediQ Health Companion – AI-Powered Healthcare Platform

- Implemented multimodal health management platform integrating Gemini 1.5 Pro for synthesizing medical image analysis with longitudinal health tracking data.
- Optimized scalable backend on FastAPI and MongoDB to manage high-throughput medical data processing while maintaining low-latency user insights.
- Designed modular AI workflow that translates raw diagnostic markers into patient-centric health summaries and preventative action plans.
- Tech:** Python, FastAPI, React, Computer Vision, Gemini API, MongoDB.

AutoReferral – AI Healthcare Referral System

- Deployed clinical agentic system during 24-hour sprint that classifies medical referrals using calibrated confidence scoring to minimize false positives.
- Integrated explainable AI layer that provides physician-ranked recommendations based on historical referral outcomes and specialist availability.
- Refined prompt-chaining architecture to ensure high accuracy and rapid inference under intensive hackathon time constraints.
- **Tech:** FastAPI, React, Agentic AI, Gemini API.

BluCarbon – Web3 Carbon Verification Platform

- Integrated Sentinel Hub satellite imagery feeds with CNN-based field verification to automate assessment of blue carbon sequestration sites.
- Developed decentralized verification pipeline on Polygon network, minting authenticated carbon credit NFTs based on AI-validated field data.
- Architected secure API gateway connecting on-chain smart contracts with off-chain geospatial processing nodes for real-time credit auditing.
- **Tech:** React, FastAPI, Solidity, Polygon, Sentinel Hub, MongoDB.

HONORS & LEADERSHIP

- **Technical Lead – Ignisia AI Hackathon**

Led rapid AI product prototyping and deployment during national-level hackathon participation.

- **Technical Core Member – CSI MIT-WPU**

Contributed to campus-wide software engineering and technical community initiatives.

Certifications:

- Machine Learning Operations (MLOps) for Generative AI – Google
- Introduction to Model Context Protocol – Anthropic
- Introduction to Subagents – Anthropic
- JPMorganChase – Software Engineering Job Simulation

EDUCATION

MIT World Peace University

B.Tech in Computer Science (Specialization in AI & Data Science)

Pune, India

Aug 2023 – June 2029